

Dynamic Pedestrian Flow Control Using Flow-Robot Interaction Model of Mobile Robot and Human Crowd

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Abstract—This study addresses congestion caused by dense populations, promoting safe, efficient, and comfortable mobility in sustainable societies. It focuses on mobile robots for crowd management, proposing a novel method using Flow-Robot Interaction (FRI) and a heuristic behavior-based Social Force Model (SFM) to dynamically control pedestrian overtaking flows. Robots are strategically positioned at overtaking points to interact effectively with pedestrians, restricting paths and minimizing trajectory disruptions. Unlike fixed obstacles, the robots can flexibly reposition to obstruct pedestrians in their desired direction. Simulations show that placing robots at overtaking points stabilizes flows and reduces trajectory changes. Real-world experiments with 11 participants validate these findings, showing alignment of flow velocity with slower pedestrians and prevention of flow expansion. Results demonstrate the potential of human-robot interaction to mitigate congestion, stabilize flows, and enhance mobility. This scalable method offers an effective solution for managing crowds in sustainable urban environments.

Keywords—flow-robot interaction, pedestrian flow control, mobile robot, motion planning

I. INTRODUCTION

Mitigating congestion in stations, event venues, and intersections helps the safe, comfortable, and efficient movement of people and contributes to the realization of a sustainable society. Crowd control technologies facilitate smooth pedestrian flow in normal conditions and enable rapid, safe evacuation during emergencies. This study proposes a method to reduce congestion and improve pedestrian flow by designing effective Flow-Robot Interaction (FRI).

Crowd control uses technology to manage congestion and flow, promoting behaviors that ensure safe and efficient movement. Common methods include placing static obstacles [1-3], applying nudge theory to leverage psychological effects [4], and providing guidance information [5-6]. However, these methods struggle to address complex and nonlinear crowd behaviors. Mobile robots offer a new approach to managing dynamic crowd behavior. Strategies for controlling crowd behavior can draw inspiration from traffic congestion management on highways, such as ramp metering, high-occupancy vehicle (HOV) lanes, and active traffic management (ATM) [7]. On highways, congestion is controlled by providing drivers with visual feedback. Similarly, crowds can be managed using visual information. However, crowd behavior is more complex due to nonlinear movements, psychological factors,

and social interactions. Therefore, it is essential for robots to directly influence pedestrian flow. Wan et al. used deep reinforcement learning for robot motion planning to control flow efficiently [8]. Yamamoto et al. developed a method to improve average velocity in crossing flows using guides [9]. Jiang et al. combined mobile robots with adaptive dynamic programming to maximize flow and reduce crowd pressure [10]. These studies highlight the potential of FRI for flow control. However, the effectiveness of robot-based crowd control in real environments remains underexplored. Robots also need to perceive the flow accurately and plan suitable movements.

Fig. 1 illustrates a scenario where the proposed pedestrian flow control method using mobile robots is applied. Among various crowd flow issues, such as collisions in crossing or opposing flows and reduced flow efficiency at bottlenecks, overtaking is identified as a representative example due to its high risk of causing accidents and significantly lowering mobility efficiency. This study evaluates and validates the effectiveness of the proposed method in addressing overtaking scenarios. Overtaking due to differences in pedestrian velocity and direction destabilizes flow. Flow stability refers to consistent velocity and minimal trajectory disruptions. This study proposes controlling overtaking flows by strategically positioning robots to maximize interaction with dynamic flows. Robots recognize dynamic flows and detect faster flows overtaking slower ones. Positioned at overtaking points, robots use the Social Force Model (SFM) [11] based on heuristic behavior to influence leading pedestrians. This method reduces velocity changes and stabilizes movement trajectories. Multiple



Fig. 1. A scenario in which mobile robots control pedestrian flow.

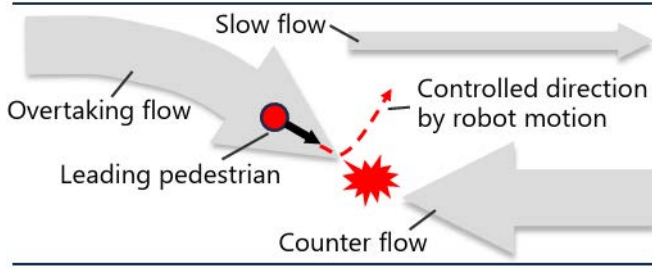


Fig. 2. Flow-Robot Interaction for overtaking flow control.

simulations and real-world experiments validate the method's effectiveness in overtaking flow scenarios. Flow velocity and trajectory variation serve as performance metrics. This research demonstrates that human-robot interaction effectively mitigates congestion, enabling safe and efficient movement in aging, urbanizing societies, and supports a sustainable future.

II. DYNAMIC PEDESTRIAN FLOW CONTROL BASED ON FLOW-ROBOT INTERACTION

This study proposes a new method to control overtaking flows using FRI and heuristic-behavior-based SFM.

A. Specification of Flow-Robot Interaction

Using FRI enables a robot to control overtaking flows. Fig. 2 shows Flow-Robot Interaction for overtaking flow control. Intersecting pedestrian flows with different velocities cause collisions, reducing safety and comfort. A robot can regulate the entire flow by controlling the desired direction of the leading pedestrian.

B. Social Force Model Based on Heuristic behavior

This study utilizes SFM based on the heuristic behavioral model proposed by Moussaïd et al. [11] to control pedestrians' desired movement directions.

The following describes the SFM based on the heuristic behavioral model. Within the field of view $[\varphi, \varphi]$ of pedestrian i , the desired direction of movement α_{des} is determined as the direction that minimizes the distance $d(\alpha)$ to the destination O_i . This direction is expressed by the following minimization problem:

$$\alpha_{des} = \operatorname{argmin}_{\alpha} d(\alpha) \quad (1)$$

$d(\alpha)$ is defined using the maximum distance d_{max} , which is determined by the pedestrian's vision, as follows:

$$d(\alpha) = d_{max}^2 + f(\alpha)^2 - 2d_{max}f(\alpha)\cos(\alpha_0 - \alpha) \quad (2)$$

Here, $f(\alpha)$ represents the distance to the first collision when pedestrian i moves in the desired direction α_{des} at their comfortable maximum speed v_0 . This value is calculated considering the speed and physical size of other pedestrians, and it reflects potential future collisions. Additionally, α_0 indicates the direction toward the destination.

The desired speed v_{des} of the pedestrian is determined by principles necessary to avoid collisions with obstacles or other pedestrians and is expressed as $v_{des} = \min(v_0, d_h/\tau)$. Here, d_h is the distance to the first obstacle in the desired direction α_{des} , and τ represents the safety time.

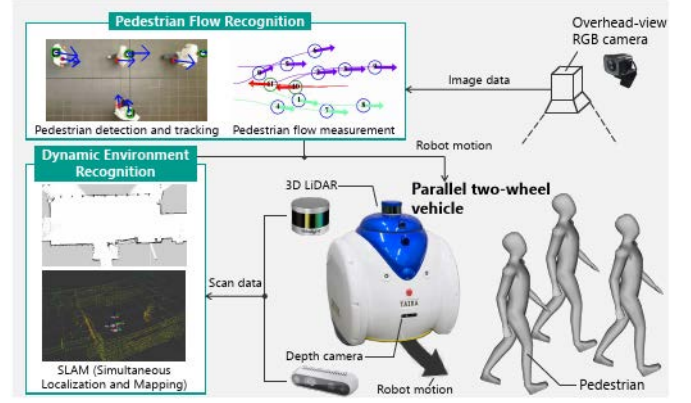


Fig. 3. Schematic of pedestrian flow and environment recognition system.

Using the desired direction α_{des} and desired speed v_{des} , the desired velocity $\mathbf{v}_{des}$ can be expressed as follows:

$$\mathbf{v}_{des} = v_{des}(\cos \alpha_{des}, \sin \alpha_{des}) \quad (3)$$

Therefore, under normal walking conditions, the change in the actual velocity $\mathbf{v}_i$ at time t is expressed by the following acceleration equation:

$$d\mathbf{v}_i/dt = (\mathbf{v}_{des} - \mathbf{v}_i)/\tau \quad (4)$$

C. Overtaking Pedestrian Flow Control Algorithm

Overtaking flow control algorithm is described. The desired direction of the leading pedestrian is calculated using SFM. Robot's velocity is determined to maximize the objective function $d(\alpha)$. By minimizing collision time while considering relative velocity and body size, a robot can effectively influence the pedestrian's direction.

III. DYNAMIC PEDESTRIAN FLOW AND ENVIRONMENT RECOGNITION SYSTEM

Fig. 3 shows the system configuration for capturing pedestrian flow and environmental information to enable robot to control human crowds. The system uses an overhead RGB camera to detect and track pedestrians' heads based on color. Pedestrians' positions and velocities are derived from detection and tracking results to measure pedestrian flow dynamics. The robot is a two-wheeled mobile platform equipped with a 3D LiDAR and a Depth camera. Point cloud data from these sensors supports environmental mapping, self-localization, and obstacle detection. The system integrates pedestrian flow and environmental data to plan robot motions. This enables the robot to perceive crowd flow accurately and intervene effectively.

IV. VERIFICATION OF FLOW CONTROLLED EFFECT BY USING MOBILE ROBOT AS DYNAMIC OBSTACLE

Multiple simulations and real-world experiments validate the effectiveness of the overtaking flow control method in overtaking scenarios.

A. Dynamic Pedestrian Flow Control Simulation

In a scenario where a faster flow (1.0 m/s) overtakes a slower flow (0.5 m/s), the effectiveness of the overtaking flow control method based on FRI is verified. A simulation is conducted under the condition where the robot is positioned at the overtaking point near the flow boundary. Fig. 4 shows that

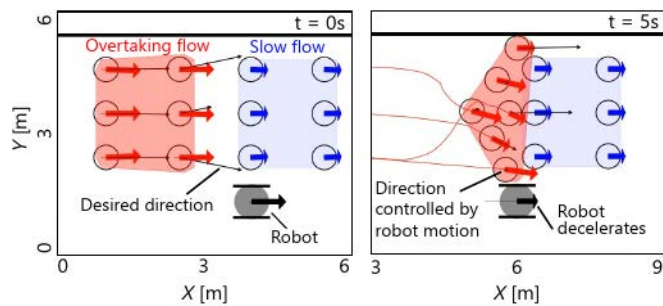


Fig. 4. Snapshots of overtaking pedestrian flow control.

positioning the robot at the overtaking point minimizes flow region changes and stabilizes flow paths. Simulation results confirm that appropriate robot positioning is critical in intersecting flows with different velocities. Controlling the leading pedestrian in the overtaking flow demonstrates that a single robot can effectively control the entire flow.

B. Experimental Pedestrian Flow Control

To verify flow control effectiveness in real-world environments, overtaking flow control experiments using remote-controlled robots are conducted. Eleven male participants, aged between 22 and 24, are involved. Informed consent is obtained from all participants in advance. The purpose, methods, and potential risks of the study are explained to the participants, and they provided their consent after understanding the information.

The control effect is evaluated under conditions with flow control (Fig. 5). In Fig. 5, where the robot controls the leading pedestrian, pedestrian overtaking is suppressed, and a flow region remains stable. These results confirm the effectiveness of mobile robot-based flow control in real-world environments. They suggest that mobile robots can be valuable tools for crowd management.

V. CONCLUSION

Mobile robots can adapt to nonlinear and dynamic crowds by recognizing pedestrian behavior and the surrounding environment in real time. This capability allows them to control dynamic crowd flows effectively and mitigate congestion. This study proposes a novel approach to managing pedestrian flows by leveraging Flow-Robot Interaction (FRI) and heuristic-behavior-based Social Force Model (SFM). A two-wheeled mobile robot equipped with 3D LiDAR and depth sensors acts as a dynamic obstacle that flexibly repositions within the flow. The method's effectiveness is evaluated through simulations and real-world experiments involving 11 participants. Results show that strategically positioning the robot at overtaking points effectively stabilizes overtaking flows. Simulations demonstrate that the method reduces trajectory disruptions and aligns faster pedestrian flows with slower ones. Real-world experiments confirm that the robot suppresses overtaking behaviors, prevents flow region expansion, and maintains stability. This research contributes to addressing urban mobility challenges by providing an effective solution for enhancing pedestrian safety, comfort, and efficiency.

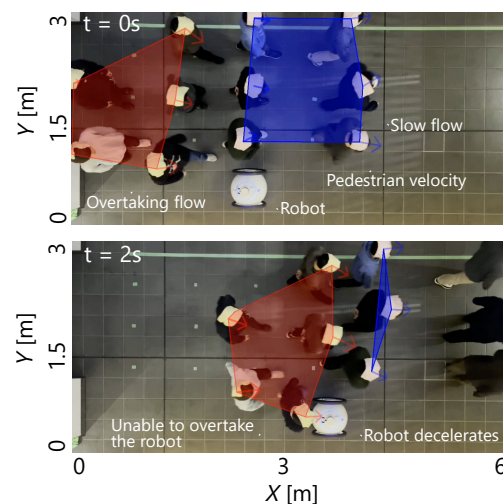


Fig. 5. Snapshots of experimental pedestrian flow control using mobile robot.

As the near future work, we intend to validate the method in diverse environments and scenes, including congested urban areas and large-scale venues. We also aim to develop multi-robot cooperative control in managing pedestrian flows.

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